

PRACTICAL- 5:

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nirvighna@DESKTOP-VS57D2  ×  +  ▾
GNU nano 7.2 producer_consumer.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];
    pid_t pid;

    char message[] = "Hello Child! This data is produced by the parent.";
    char buffer[100];

    if (pipe(pipefd) == -1)
    {
        perror("pipe failed");
        return 1;
    }

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid > 0)
    {
        // Parent - Producer
        close(pipefd[0]);

        printf("Parent (Producer): Generating data...\n");
        printf("Parent: Sending message to child...\n");

        write(pipefd[1], message, strlen(message) + 1);
    }

    if (pid < 0)
    {
        // Child - Consumer
        read(pipefd[0], buffer, 100);

        printf("Child (Consumer): Data received from parent: %s\n", buffer);
        printf("Child: Hello Child! This data is produced by the parent.\n");
        printf("Child: Consumer completed.\n");
    }

    printf("Parent: Producer completed.\n");
    return 0;
}

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify

nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./producer_consumer
Parent (Producer): Generating data...
Parent: Sending message to child...
Child (Consumer): Data received from parent:
Child: Hello Child! This data is produced by the parent.
Child: Consumer completed.
Parent: Producer completed.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
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nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./producer_consumer
Parent (Producer): Generating data...
Parent: Sending message to child...
Child (Consumer): Data received from parent:
Child: Hello Child! This data is produced by the parent.
Child: Consumer completed.
Parent: Producer completed.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l | grep ".c"
-rw-r--r-- 1 nirvighna nirvighna 1940 Sep 21 04:00 mini_shell.c
-rwxr-xr-x 1 nirvighna nirvighna 16440 Sep 25 18:02 producer_consumer
-rw-r--r-- 1 nirvighna nirvighna 1133 Sep 25 18:02 producer_consumer.c
drwxr-xr-x 2 nirvighna nirvighna 4096 Sep 25 17:21 src
-rw-r--r-- 1 nirvighna nirvighna 746 Sep 25 17:34 strace_output.txt
-rw-r--r-- 1 nirvighna nirvighna 1382 Sep 25 17:47 wait_demo.c
-rw-r--r-- 1 nirvighna nirvighna 625 Sep 25 17:51 zombie.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l
total 96
-rw-r--r-- 1 nirvighna nirvighna 498 Sep 21 05:14 Makefile
drwxr-xr-x 2 nirvighna nirvighna 4096 Sep 25 17:22 bin
-rw-r--r-- 1 nirvighna nirvighna 54 Sep 25 15:03 input.txt
-rw-r--r-- 1 nirvighna nirvighna 1940 Sep 21 04:00 mini_shell.c
-rw----- 1 nirvighna nirvighna 1193 Sep 25 13:34 nano.2750.save
-rw-r--r-- 1 nirvighna nirvighna 54 Sep 25 17:34 output.txt
-rwxr-xr-x 1 nirvighna nirvighna 16440 Sep 25 18:02 producer_consumer
-rw-r--r-- 1 nirvighna nirvighna 1133 Sep 25 18:02 producer_consumer.c
drwxr-xr-x 2 nirvighna nirvighna 4096 Sep 25 17:21 src
-rw-r--r-- 1 nirvighna nirvighna 746 Sep 25 17:34 strace_output.txt
-rwxr-xr-x 1 nirvighna nirvighna 16352 Sep 25 17:48 wait_demo
-rw-r--r-- 1 nirvighna nirvighna 1382 Sep 25 17:47 wait_demo.c
-rwxr-xr-x 1 nirvighna nirvighna 16216 Sep 25 17:52 zombie
-rw-r--r-- 1 nirvighna nirvighna 625 Sep 25 17:51 zombie.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ grep ".c"

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